The higher-dimensional Hardy deficit is not a norm

A negative answer to the explicit norm subquestion in arXiv:1909.12587

Automated mathematical-discovery candidate (lane 11)

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Status. Candidate counterexample; current verdict **likely valid**. The result completely answers the paper's explicit norm subquestion for every $n \geq 3$, but it does *not* settle existence of extremals for the singular Hardy–Moser–Trudinger inequality.

1 Source question and scope

Nguyen's paper [1] proves the sharp singular Hardy–Moser–Trudinger inequality on $\mathbb{B}^n$. On PDF page 5 it asks whether extremals exist when $n \geq 3$ and identifies a concrete obstruction: it was not known whether $u \mapsto \mathcal{H}(u)^{1/n}$ is a norm on $C_c^\infty(\mathbb{B}^n)$, where

$$\mathcal{H}(u) = \int_{\mathbb{B}^n} |\nabla u|^n \, dx - c_n \int_{\mathbb{B}^n} \frac{|u|^n}{(1-|x|^2)^n} \, dx, \qquad c_n = \left(\frac{2(n-1)}{n} \right)^n. \tag{1}$$

The same page proposes the analogous distance-to-the-boundary deficit on convex domains. The source passage is reproduced below.

inequality. For the improved Moser–Trudinger inequality, the existence of extremals was proved in [35, 37, 54] and the references therein. In [19, 20], Li developed a blow-up analysis method to establish the existence of extremals for the Moser–Trudinger inequality on Riemannian manifolds. Concerning to the Hardy–Moser–Trudinger inequality, it was proved by Wang and Ye [47] (by using the blow-up analysis method) that the extremals for the inequality (1.6) exists in $\mathcal{H}(\mathbb{B}^2)$ but not in $W_0^{1,2}(\mathbb{B}^2)$. Similarly, again by the the blow-up analysis method, Yang and Zhu proved the existence of extremals for the improvement version of (1.6) and Wang proved the existence of the singular Hardy–Moser–Trudinger inequality (1.9) in $\mathbb{B}^2$. It remains an open question in this paper which is whether or not the extremals for the singular Hardy–Moser–Trudinger inequality (1.9) exists when $n \geq 3$. The main difficult is to determine the suitable space for which the extremals (if exist) belong to. Let us recall that when $n \geq 3$, we do not know the functional $u \to \mathcal{H}(u)^{\frac{1}{n}}$ is a norm on $C_0^\infty(\mathbb{B}^n)$ or not. So we can't talk about the completion of $C_0^\infty(\mathbb{B}^n)$ under this functional also the weak convergence with respect to this functional. This is the crucial different with the case $n = 2$. We will come back this question in the future research.

As a final remark, it is well known that for a convex domain domain $\Omega \subset \mathbb{R}^n$ the following Hardy's inequality holds (see, e.g., [28, 30])

$$\int_\Omega |\nabla u|^n dx \geq \left(\frac{n-1}{n}\right)^n \int_\Omega \frac{|u|^n}{d(x,\partial\Omega)^n} dx, \quad u \in C_0^\infty(\Omega),$$

where $d(x,\partial\Omega) = \inf\{|x-y| : y \in \partial\Omega$. The constant $(n-1)^n/n^n$ is sharp and never attained. Hence,

$$H_\Omega(u) = \int_\Omega |\nabla u|^n dx - \left(\frac{n-1}{n}\right)^n \int_\Omega \frac{|u|^n}{d(x,\partial\Omega)^n} dx > 0, \quad u \in C_0^\infty(\Omega) \setminus \{0\}.$$

We wonder if the inequality (1.9) can be extended to any convex domain Ω in $\mathbb{R}^n$. In this direction, we propose the following inequality

$$\sup_{u \in C_0^\infty(\Omega), H_\Omega(u) \leq 1} \int_\Omega e^{\alpha_n (1-\frac{\beta}{n})|u|^{\frac{n}{n-1}}} |x|^{-\beta} dx < \infty. \tag{1.11}$$

Since $d(x,\partial\mathbb{B}^n) = 1 - |x| \geq \frac{1-|x|^2}{2}$, then the inequality (1.11) holds when $\Omega = \mathbb{B}^n$ by (1.9). In dimension two, the inequality (1.11) for $\beta = 0$ was conjectured by Wang and Ye (see [47, Conjecture, page 4]) and was recently settled by Lu and Yang [24].

The rest of this paper is organized as follows. In the section §2 we recall some facts on the rearrangement arguments in the hyperbolic space which enables us reducing the proof of (1.9) to the radial functions in $\mathbb{B}^n$. We also prove the existence of the Green function G and its properties in this section. Finally, we define a transformation of functions (based on the transplantation of Green functions) and make some useful computations which is useful in the proof of(1.9) in subsection §2.3. The proofs of (1.9) and (1.10) are given in the section §3. We also make some further comments on the application of our method to obtain the other improvements of the

The result below answers the norm subquestion negatively. It also shows that the same obstruction persists for the source's natural smooth-convex-domain functional. The larger extremal-existence question remains open.

2 Proof intuition

If $N = \mathcal{H}^{1/n}$ were a norm, then $\mathcal{H} = N^n$ would be convex. We disprove convexity by looking at the second variation in a thin boundary annulus. On that annulus choose a base function which agrees with $1 - |x|^2$. After the normal coordinate $s = 1 - |x|$ is introduced, its second variation is, up to a positive asymptotically constant factor,

$$\int \left(|h'|^2 - \kappa_n \frac{|h|^2}{s^2}\right) ds, \qquad \kappa_n = \left(\frac{n-1}{n}\right)^n.$$

Smooth logarithmic boundary layers make the quotient of the two integrals tend to the one-dimensional Hardy constant $1/4$. But $\kappa_n > 1/4$ precisely throughout the requested range $n \geq 3$. Thus the Hessian has a negative direction, which is incompatible with any norm. In a smooth domain, Fermi coordinates give the same normal expression; curvature and tangential terms are lower-order errors.

3 A smooth logarithmic boundary layer

Lemma 1. *For every $\delta > 0$ there are nonzero $h_L \in C_c^\infty(0,\delta)$ such that*

$$\frac{\int_0^\delta |h_L'(s)|^2\,ds}{\int_0^\delta |h_L(s)|^2 s^{-2}\,ds} \longrightarrow \frac{1}{4}. \tag{2}$$

Proof. Fix nonzero $\chi \in C_c^\infty(0,1)$ and put $\varepsilon_L = \delta e^{-L}$, $g_L(t) = \chi(t/L)$, and

$$h_L(s) = s^{1/2} g_L\left(\log \frac{s}{\varepsilon_L}\right).$$

Because χ vanishes in neighborhoods of both endpoints, h_L is smooth and compactly supported in (ε_L, δ). With $t = \log(s/\varepsilon_L)$,

$$D_L := \int_0^\delta \frac{h_L^2}{s^2}\,ds = \int_0^L g_L^2\,dt = L\int_0^1 \chi^2\,dt,$$

$$\int_0^\delta |h_L'|^2\,ds = \int_0^L \left(g_L' + \tfrac{1}{2}g_L\right)^2 dt = \tfrac{1}{4}D_L + \int_0^L |g_L'|^2 dt.$$

The cross term vanishes by compact support, while $\int |g_L'|^2 = L^{-1}\int |\chi'|^2$. Hence the quotient in (2) equals

$$\frac{1}{4} + \frac{1}{L^2}\frac{\int_0^1 |\chi'|^2}{\int_0^1 |\chi|^2},$$

which proves the claim. ◻

4 Negative answer on the ball

Theorem 1. *Let $n \geq 3$. The functional $\mathcal{H}$ in (1) is not convex on $C_c^\infty(\mathbb{B}^n)$. Consequently $u \mapsto \mathcal{H}(u)^{1/n}$ is not a norm on $C_c^\infty(\mathbb{B}^n)$.*

Proof. Set $\kappa_n = ((n-1)/n)^n$. The sequence $(1-1/n)^n$ is increasing for $n \geq 2$: for real $x > 1$, the derivative of $x\log(1-1/x)$ is $\log(1-1/x) + 1/(x-1) > 0$, by $\log(1+y) < y$. Hence

$$\kappa_n \geq \left(\frac{2}{3}\right)^3 = \frac{8}{27} > \frac{1}{4}. \tag{3}$$

We first calculate with a radial direction $V(r)$ supported in a sufficiently thin annulus next to $r = 1$. Choose $U \in C_c^\infty(\mathbb{B}^n)$ radial and equal to $1 - r^2$ on a neighborhood of $\operatorname{supp} V$. Such a U exists because each fixed support is a positive distance from the boundary. On that support, $U > 0$ and $U' = -2r \neq 0$. Therefore differentiation under the integral is legitimate and, writing $\omega_{n-1} = |S^{n-1}|$,

$$\frac{D^2\mathcal{H}(U)[V,V]}{n(n-1)\omega_{n-1}} = \int_0^1 \left[(2r)^{n-2}(V')^2 - c_n\frac{V^2}{(1-r^2)^2}\right] r^{n-1}\,dr. \tag{4}$$

Indeed, the scalar identity $\frac{d^2}{dt^2}|a+tb|^n|_{t=0} = n(n-1)|a|^{n-2}b^2$ applies to both radial terms.

Put $s = 1 - r$ and $h(s) = V(1-s)$. The right-hand side of (4) becomes

$$\int_0^1 \left[A_n(s)|h'|^2 - C_n(s)\frac{|h|^2}{s^2}\right] ds, \tag{5}$$

where

$$A_n(s) = 2^{n-2}(1-s)^{2n-3}, \qquad C_n(s) = c_n \frac{(1-s)^{n-1}}{(2-s)^2}.$$

As $s \downarrow 0$,

$$A_n(s) \longrightarrow 2^{n-2}, \qquad C_n(s) \longrightarrow \frac{c_n}{4} = 2^{n-2}\kappa_n.$$

Because of the strict gap in (3), choose $\eta > 0$ so small that $(1+\eta)(1/4+\eta) < \kappa_n - \eta$. Then choose $\delta > 0$ so that on $(0, \delta)$,

$$A_n(s) \leq 2^{n-2}(1+\eta), \qquad C_n(s) \geq 2^{n-2}(\kappa_n - \eta).$$

Finally take $h = h_L$ from Lemma 1 with L large enough that $\int |h'|^2 < (1/4+\eta) \int h^2/s^2$. Equation (5) is then strictly negative. Defining $V(r) = h(1-r)$ gives a member of $C_c^\infty(\mathbb{B}^n)$, and the compactly supported U described above gives

$$D^2\mathcal{H}(U)[V,V] < 0. \tag{6}$$

For sufficiently small $t > 0$, Taylor's theorem and (6) give

$$\mathcal{H}(U+tV) + \mathcal{H}(U-tV) - 2\mathcal{H}(U) < 0,$$

which violates midpoint convexity. If $N = \mathcal{H}^{1/n}$ were a norm, then

$$\mathcal{H}(\theta f + (1-\theta)g) = N(\theta f + (1-\theta)g)^n \leq \big(\theta N(f) + (1-\theta)N(g)\big)^n \leq \theta\mathcal{H}(f) + (1-\theta)\mathcal{H}(g),$$

so $\mathcal{H} = N^n$ would be convex. This contradiction proves that N is not a norm. □

5 Structure-preserving domain extension

For a bounded smooth convex domain $\Omega \subset \mathbb{R}^n$, let

$$\mathcal{H}_\Omega(u) = \int_\Omega |\nabla u|^n \, dx - \kappa_n \int_\Omega \frac{|u|^n}{d(x,\partial\Omega)^n} \, dx. \tag{7}$$

The strict Hardy inequality quoted in the source makes $\mathcal{H}_\Omega(u)^{1/n}$ pointwise well-defined on nonzero test functions.

Corollary 1. *For every bounded C^∞ convex domain $\Omega \subset \mathbb{R}^n$ and every $n \geq 3$, $\mathcal{H}_\Omega$ is nonconvex on $C_c^\infty(\Omega)$. Hence $\mathcal{H}_\Omega^{1/n}$ is not a norm.*

Proof. Use Fermi coordinates (y, s) in a smooth boundary patch, with $s = d(x, \partial\Omega)$. Choose $U \in C_c^\infty(\Omega)$ which equals s on a neighborhood of the support of the direction below. For $V(y,s) = \zeta(y)h_L(s)$, where ζ is a fixed nonzero smooth tangential cutoff, the same differentiation as above yields

$$\frac{D^2\mathcal{H}_\Omega(U)[V,V]}{n(n-1)} = \int \left[(\partial_s V)^2 + \frac{1}{n-1}|\nabla_{T,s}V|^2 - \kappa_n \frac{V^2}{s^2}\right] J(y,s)\, dy\, ds. \tag{8}$$

Here $J(y,s) = J(y,0)(1+O(s))$, and the tangential metrics are uniformly equivalent for small s. Let $D_L = \int h_L^2/s^2$. Lemma 1 gives $\int |h_L'|^2 = (1/4 + o_L(1))D_L$, while

$$\int_0^\delta h_L^2 \, ds \leq \delta^2 D_L.$$

Consequently the Jacobian/metric errors in the normal and potential terms are $O(\delta)D_L$, and the nonnegative tangential term is $O(\delta^2)D_L$. First choose δ so small that these errors are less than half the strict gap $\kappa_n - 1/4$, and then choose L large. The expression in (8) is negative. Thus $\mathcal{H}_\Omega$ is nonconvex, and the same norm-implies-convexity argument as in Theorem 1 finishes the proof. □

6 Verification, limitations, and novelty status

The proof is analytic and has no computational dependency. Verification checked: (i) the exact second derivative factors; (ii) the limit $c_n/(4 \cdot 2^{n-2}) = \kappa_n$; (iii) the strict threshold $\kappa_n > 1/4$ for all $n \geq 3$; (iv) genuine smooth compact support of every fixed logarithmic layer and its localized base; and (v) the order of the Fermi- coordinate errors. No contradiction was found.

This packet does *not* prove either existence or nonexistence of an extremal for the singular inequality. It shows instead that the particular norm-completion route identified in the source cannot work: the proposed gauge already fails the triangle inequality on test functions. Likewise, Corollary 1 does not decide the source's exponential inequality on convex domains.

A bounded search on 12 August 2026 covered the exact title and arXiv id, the quoted norm question, `Hardy--Moser--Trudinger extremals n>=3`, `Hardy deficit functional convexity p`, and `Hardy functional does not define a norm`. It found the source, dimension-two results, and general Hardy papers, but no answer to this exact higher-dimensional norm question and no matching nonconvexity theorem. Novelty is therefore plausible, not certified. Recommended human-review focus is the compact-support localization in Theorem 1 and the error bookkeeping in Corollary 1.

References

[1] V. H. Nguyen, *The sharp Hardy–Moser–Trudinger inequality in dimension n*, arXiv:1909.12587 (2019), especially Theorem 1.1 and PDF page 5.